

6.0 TIMER0 MODULE

The Timer0 module timer/counter has the following features:

- 8-bit timer/counter
- Read/write capabilities
- 8-bit software programmable prescaler
- Internal or external clock select
- Interrupt on overflow from FFh to 00h
- Edge select for external clock

Figure 6-1 is a simplified block diagram of the Timer0 module. Additional information is available in the “PIC® Mid-Range MCU Family Reference Manual” (DS33023).

Timer mode is selected by clearing the T0CS bit (OPTION<5>). In Timer mode, the TMR0 register value will increment every instruction cycle (without prescaler). If the TMR0 register is written to, the increment is inhibited for the following two cycles. The user can work around this by writing an adjusted value to the TMR0 register.

Counter mode is selected by setting the T0CS bit. In this mode the TMR0 register value will increment either on every rising or falling edge of pin RA4/T0CKI/CMP2. The incrementing edge is determined by the source edge (T0SE) control bit (OPTION<4>). Clearing the T0SE bit selects the rising edge. Restrictions on the external clock input are discussed in detail in **Section 6.2 “Using Timer0 with External Clock”**.

The prescaler is shared between the Timer0 module and the Watchdog Timer. The prescaler assignment is controlled in software by the control bit PSA (OPTION<3>). Clearing the PSA bit will assign the prescaler to Timer0. The prescaler is not readable or writable. When the prescaler is assigned to the Timer0 module, prescale value of 1:2, 1:4,..., 1:256 are selectable. **Section 6.3 “Timer0 Prescaler”** details the operation of the prescaler.

6.1 Timer0 Interrupt

Timer0 interrupt is generated when the TMR0 register timer/counter overflows from FFh to 00h. This overflow sets the T0IF bit. The interrupt can be masked by clearing the T0IE bit (INTCON<5>). The T0IF bit (INTCON<2>) must be cleared in software by the Timer0 module interrupt service routine before re-enabling this interrupt. The Timer0 interrupt cannot wake the processor from Sleep since the timer is shut off during Sleep.

6.2 Using Timer0 with External Clock

When an external clock input is used for Timer0, it must meet certain requirements. The external clock requirement is due to internal phase clock (Tosc) synchronization. Also, there is a delay in the actual incrementing of Timer0 after synchronization.

6.2.1 EXTERNAL CLOCK SYNCHRONIZATION

When no prescaler is used, the external clock input is the same as the prescaler output. The synchronization of T0CKI with the internal phase clocks is accomplished by sampling the prescaler output on the Q2 and Q4 cycles of the internal phase clocks (Figure 6-1). Therefore, it is necessary for T0CKI to be high for at least 2Tosc (and a small RC delay of 20 ns) and low for at least 2Tosc (and a small RC delay of 20 ns). Refer to the electrical specification of the desired device.

When a prescaler is used, the external clock input is divided by the asynchronous ripple-counter type prescaler so that the prescaler output is symmetrical. For the external clock to meet the sampling requirement, the ripple-counter must be taken into account. Therefore, it is necessary for T0CKI to have a period of at least 4Tosc (and a small RC delay of 40 ns) divided by the prescaler value. The only requirement on T0CKI high and low time is that they do not violate the minimum pulse width requirement of 10 ns. Refer to parameters 40, 41 and 42 in the electrical specification of the desired device. See Table 17-8.

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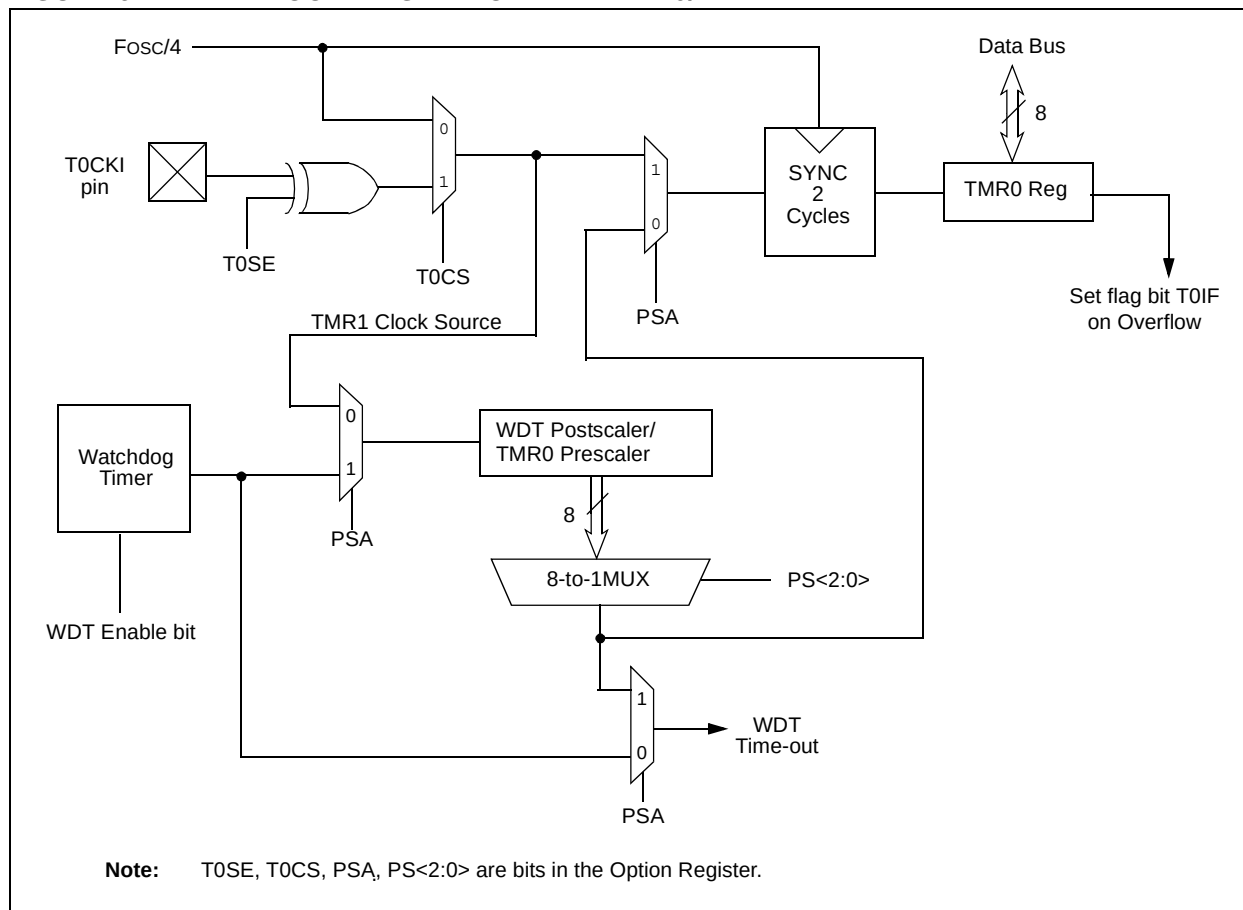
6.3 Timer0 Prescaler

An 8-bit counter is available as a prescaler for the Timer0 module, or as a postscaler for the Watchdog Timer. A prescaler assignment for the Timer0 module means that there is no postscaler for the Watchdog Timer, and vice-versa.

The PSA and PS<2:0> bits (OPTION<3:0>) determine the prescaler assignment and prescale ratio.

When assigned to the Timer0 module, all instructions writing to the TMR0 register (e.g., `CLRF 1`, `MOVWF 1`, `BSF 1, x...` etc.) will clear the prescaler. When assigned to WDT, a `CLRWDT` instruction will clear the prescaler along with the Watchdog Timer. The prescaler is not readable or writable.

FIGURE 6-1: BLOCK DIAGRAM OF THE TIMER0/WDT



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6.3.1 SWITCHING PRESCALER ASSIGNMENT

The prescaler assignment is fully under software control (i.e., it can be changed “on-the-fly” during program execution). Use the instruction sequences shown in Example 6-1 when changing the prescaler assignment from Timer0 to WDT, to avoid an unintended device Reset.

EXAMPLE 6-1: CHANGING PRESCALER (TIMER0 → WDT)

```
BCF     STATUS, RP0    ;Skip if already in
                        ;Bank 0
CLRWDT                      ;Clear WDT
CLRF     TMR0           ;Clear TMR0 and
                        ;Prescaler
BSF     STATUS, RP0    ;Bank 1
MOVLW   '00101111'b    ;These 3 lines
                        ;(5, 6, 7)
MOVWF   OPTION_REG     ;are required only
                        ;if desired PS<2:0>
                        ;are
CLRWDT                      ;000 or 001
MOVLW   '00101xxx'b    ;Set Postscaler to
MOVWF   OPTION_REG     ;desired WDT rate
BCF     STATUS, RP0    ;Return to Bank 0
```

To change prescaler from the WDT to the Timer0 module, use the sequence shown in Example 6-2. This precaution must be taken even if the WDT is disabled.

EXAMPLE 6-2: CHANGING PRESCALER (WDT → TIMER0)

```
CLRWDT                      ;Clear WDT and
                        ;prescaler
BSF     STATUS, RP0
MOVLW   b'xxxx0xxx'      ;Select TMR0, new
                        ;prescale value and
                        ;clock source
MOVWF   OPTION_REG
BCF     STATUS, RP0
```

TABLE 6-1: REGISTERS ASSOCIATED WITH TIMER0

Address	Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Value on POR	Value on All Other Resets
01h, 101h	TMR0	Timer0 Module Register								xxxx xxxx	uuuu uuuu
0Bh, 8Bh, 10Bh, 18Bh	INTCON	GIE	PEIE	TOIE	INTE	RBIE	TOIF	INTF	RBF	0000 000x	0000 000u
81h, 181h	OPTION ⁽²⁾	$\overline{\text{RBPU}}$	INTEDG	T0CS	T0SE	PSA	PS2	PS1	PS0	1111 1111	1111 1111
85h	TRISA	TRISA7	TRISA6	TRISA5	TRISA4	TRISA3	TRISA2	TRISA1	TRISA0	1111 1111	1111 1111

Legend: - = Unimplemented locations, read as '0', u = unchanged, x = unknown. Shaded cells are not used for Timer0.

Note 1: Option is referred by OPTION_REG in MPLAB® IDE Software.